Method	Memory	Budget	Mean scan time	MAPE % [95% CI]	Scans < 5%	Worst scan MAPE %
RL	**TI histogram**	**560 s**	**541 s**	**2.93 [2.58, 3.30]**	**95.8%**	**5.13**
RL	none	560 s	548 s	4.16 [3.41, 4.99]	66.7%	9.50
RL	LSTM[†]	560 s	525 s	7.31 [6.10, 8.65]	25.0%	15.02
Fixed, distilled from RL policy[§]	–	560 s	549 s	3.34 [2.30, 4.86]	91.7%	17.87
Fixed log TI grid	–	560 s	504 s	6.04 [5.36, 6.70]	25.0%	10.29
Cramér-Rao optimal	–	560 s	558 s	7.74 [7.08, 8.42]	8.3%	11.72
RL	none[‡]	240 s	227 s	4.62 [4.06, 5.19]	62.5%	7.74
Fixed log TI grid	–	240 s	163 s	6.86 [5.83, 7.90]	20.8%	12.04
Cramér-Rao optimal	–	240 s	240 s	15.96 [12.26, 19.81]	0.0%	35.53

[†] LSTM: selected checkpoint came from the cached rung, before any full-Bloch training; the run had 7.3× fewer full-Bloch steps than the TI-histogram run (37,862 vs 274,537).

[‡] Policy also chose the flip angle.

[§] One fixed schedule: per-block median TI and TR of the TI-histogram policy's schedules on 24 validation scans (13 blocks, 549 s).